

PLC-Based Automatic Sorting System

Project Report

Sorting of products by color and weight using vision and weight sensing

1. Introduction

Sorting is one of the most repetitive and time-consuming operations in production and packaging lines. In industries such as food processing, logistics and general manufacturing, products coming off a line usually need to be separated according to one or more physical properties before they can be packed, stored or shipped. Doing this manually is slow, inconsistent and expensive in terms of labor, which is why most modern facilities rely on automated sorting stations built around a PLC, a set of sensors and a conveyor network.

This project implements a small-scale version of such a system. The objective was not just to draw the idea on paper, but to actually build the logic and test it inside a realistic simulation environment, so that the behavior of the sensors, the conveyors and the pivot arms could be verified before any real hardware is involved. The simulation was carried out using FactoryIO, which made it possible to model the conveyor layout, the sensors and the actuators and connect them to the PLC logic exactly as they would be connected on a real line.

2. Objective of the System

The goal of the project is to design a sorting station capable of receiving a mixed stream of boxes on a single infeed conveyor and automatically routing each box to the correct output lane based on two criteria:

- Color of the box, detected using a vision sensor.
- Weight of the box, for the boxes that do not need to be separated by color.

The system is meant to demonstrate how a single PLC can combine information coming from more than one type of sensor and use it to control several actuators (pivot arms and conveyor motors) in a coordinated and energy-efficient way.

3. System Components

The station is built around the following hardware elements, all of which appear in the simulation layout shown in Figure 1:

Component	Function
Vision sensor	Mounted above the main conveyor; identifies the color of each box and reports it to the PLC.

Component	Function
Diffuse photoelectric sensor	Placed at the end of the main line to detect the two box sizes that are not sorted by color, allowing them to pass straight through.
Capacitive sensors	Positioned on the branch conveyors to detect the arrival and the passage of a box, used to time the conveyor motors and to reset the pivot arms.
Weight (load cell) sensor	Reads the weight of the grey boxes after they leave the vision station and feeds this value to the PLC for the final distribution step.
Pivot arms	Rotating deflector arms that push a box from the main conveyor onto the branch line that matches its color or weight class.
Belt conveyors (2m / 6m)	Carry the boxes between stations; each branch conveyor is switched on only while a box needs to travel on it.
PLC	Central controller that reads all sensor inputs and drives the motors and pivot arms according to the sorting logic described in Section 4.

4. Working Principle

The boxes enter the system on the main infeed conveyor in a random, mixed sequence. Two of the box types used in this layout are not separated by color, only by size: these two pass under the vision sensor without triggering a color match and continue straight ahead until they reach the diffuse sensor at the end of the line, which simply confirms that they have passed through and lets them carry on to their own outfeed.

The remaining boxes are colored green, blue or grey, and these are the ones that the vision sensor is actually looking for:

1. As a box passes under the vision sensor, the PLC reads the detected color.
2. If the color matches green or blue, the PLC immediately energizes the pivot arm assigned to that color. The arm rotates into the conveyor path and deflects the box onto its dedicated branch conveyor.
3. A capacitive sensor mounted on that branch conveyor detects the box as it moves past. Once the box has fully cleared the sensor, the PLC switches the pivot arm back to its rest position, ready for the next box, and stops the branch conveyor motor.
4. If the color is grey, the box is not diverted at this first station. Instead it continues to the weight sensor located further down the line.
5. The weight sensor measures the mass of the grey box, and depending on the value read, the PLC routes it through a second pivot arm onto one of the weight-based outfeed lanes.

Two ideas were kept central to the logic while building it. The first is that every actuator should only run when it is actually needed: a branch conveyor motor turns on the moment a box is deflected onto it and turns off again as soon as the capacitive sensor confirms the box has left, instead of running continuously. The second is that every pivot arm returns to its default position automatically once its job is done, so the next box arriving at the junction always finds the system in a known, ready state. Together these two

behaviors are what keep the station both power-efficient and reliable, since the PLC is never guessing the state of an actuator; it always knows it from a sensor reading.

5. Simulation Setup

The full layout, including the conveyors, the pivot arms, the vision sensor, the diffuse sensor, the capacitive sensors and the load cell, was built and tested in FactoryIO. This made it possible to check the timing of every conveyor and every pivot arm against the simulated boxes before writing the PLC logic, and to confirm that boxes of different colors and weights were consistently routed to the correct outfeed lane. The image below shows the simulated station with the main infeed line, the color sorting junction with its pivot arms, the weight station and the output lanes.

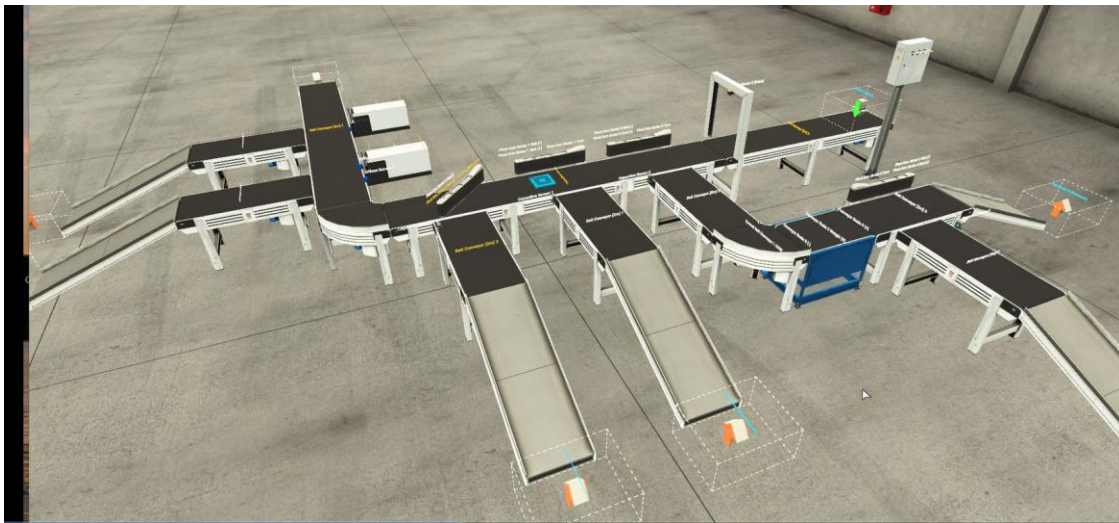


Figure 1 — Simulated layout of the automatic sorting station in FactoryIO

Running the layout in simulation before touching real hardware made it much easier to catch logic mistakes early, for example a pivot arm that did not return to rest in time for the next box, or a branch conveyor that stayed on longer than necessary. Each of these issues was visible immediately in the simulation and could be corrected in the PLC program without any risk to physical equipment.

6. Advantages of the System

- Combines two sorting criteria, color and weight, in a single station instead of requiring two separate machines.
- Branch conveyors run only when carrying a box, which reduces motor wear and saves energy compared to conveyors that run continuously.
- Pivot arms reset automatically after each box, so the station does not need to be re-armed manually between cycles.
- The layout is modular: an extra branch, sensor or color class can be added without redesigning the whole line.

- Testing in simulation first removes most of the risk and cost normally associated with commissioning a new sorting line.

7. Tools Used

The project relied on two main tools working together:

- FactoryIO — used to build the physical layout of the line: the conveyors, the pivot arms, the sensors and the load cell, and to give a realistic 3D representation of how boxes move and interact with each piece of equipment.
- PLC programming software — used to write the ladder logic that reads the sensor tags exposed by FactoryIO and drives the conveyor motors and pivot arm outputs, connected to the simulation through the standard FactoryIO I/O driver.

Working with the two together meant that every change made to the logic could be tested immediately against a moving simulation, instead of waiting until real hardware was available. This shortened the development cycle considerably and made it possible to try out different timing values for the conveyors and the pivot arms until the sorting was reliable for every box type.

8. Challenges and Adjustments

A few practical issues came up while tuning the logic, and dealing with them is part of what shaped the final design:

- Pivot arm timing: in early tests, an arm would sometimes start returning to rest before the box had fully cleared it, causing the box to be only partially deflected. This was solved by tying the return motion to the capacitive sensor on the branch conveyor rather than to a fixed delay.
- Conveyor overlap: when two boxes arrived close together, a branch conveyor could be commanded on and off too quickly. Adding a short confirmation window before switching a motor off fixed this without slowing down the overall throughput.
- Weight readings: the load cell value needed to settle for a brief period before being read, otherwise a box passing at speed could trigger a reading taken mid-transit. A short stabilization step was added before the PLC accepts the weight value for the grey boxes.

None of these issues would have been obvious from a written description of the system alone; they only appeared once the logic was actually run against moving boxes in the simulation, which is exactly why the simulation stage was treated as a required part of the project rather than an optional extra.

9. Possible Improvements

- Add a small HMI screen or web dashboard showing a running count of boxes sorted per lane, which would also make it easier to spot if one branch is jamming more often than the others.
- Log sorting data over time (color distribution, weight ranges) to support quality control reporting in a real production line.

- Add a reject lane for boxes that the vision sensor cannot classify with confidence, instead of letting them continue down the main line by default.
- Extend remote monitoring so the line status and any faults can be checked from a phone or tablet without needing to be on the factory floor.

10. Conclusion

This project shows how a relatively small set of sensors, conveyors and pivot arms can be coordinated by a single PLC to perform a sorting task that would otherwise need to be done by hand. By validating the entire sequence inside FactoryIO before any physical build, the logic for color detection, weight measurement, conveyor timing and pivot arm control was confirmed to work correctly under different combinations of incoming boxes. The same logic developed here can be transferred to a real installation with minimal changes, since the sensor and actuator behavior in the simulation was modeled to match what would be expected from real industrial components.